

Byzantine Robustness Comparison up to $f = 8$: CNN vs. SNN Encodings (Complete Sweeps)

ByzFL SNN Benchmark

June 23, 2026

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1 Introduction

This document presents comparison grids (2x2 layout) of Byzantine robustness results for CNN and SNN architectures under different data encoding schemes, extended up to $f = 8$ Byzantine clients. The configurations compared are:

1. **CNN**: Convolutional Neural Network on MNIST ('cnn_complete_plots_extended_f8')
2. **SNN Rate**: Spiking Neural Network with Rate encoding on MNIST ('snn_complete_plots_rate_extended_f8')
3. **SNN Direct**: Spiking Neural Network with Direct encoding on MNIST ('snn_complete_plots_direct_extended_f8')
4. **SNN NMNIST**: Spiking Neural Network on neuromorphic NMNIST ('snn_complete_plots_nmnist_extended_f8')

Each grid illustrates the test accuracy under varying numbers of Byzantine clients ($f \in [0, 2, 4, 6, 8]$) and data heterogeneity levels ($\gamma \in [0.0, 0.33, 0.66, 1.0]$).

2 Best Overall (Worst-Case Across All Attacks)

This section compares the performance under the worst-case attack scenario (minimizing the maximum damage over all attacks), showing both the dynamically selected best aggregator and each of the fixed aggregators.

2.1 Best Aggregator Choice

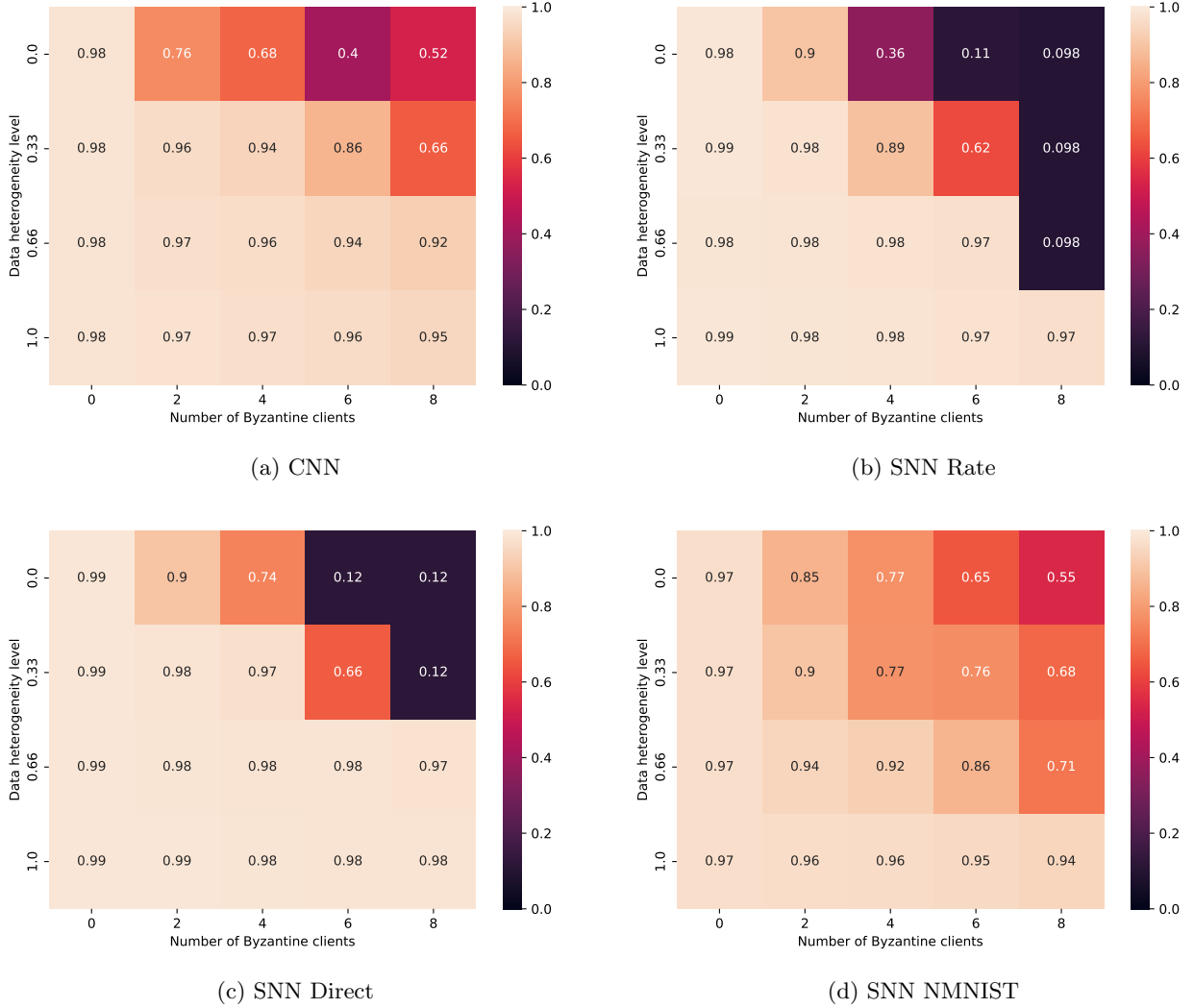
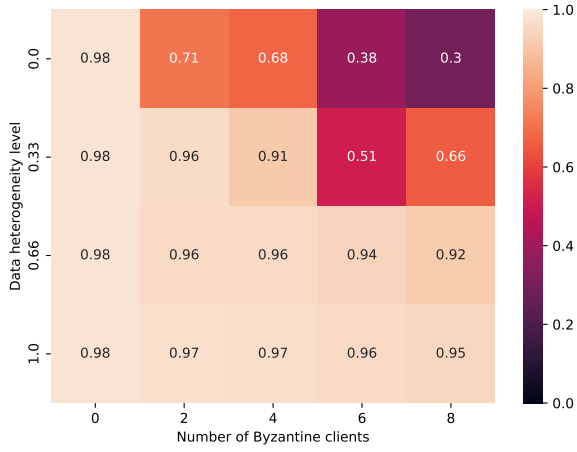
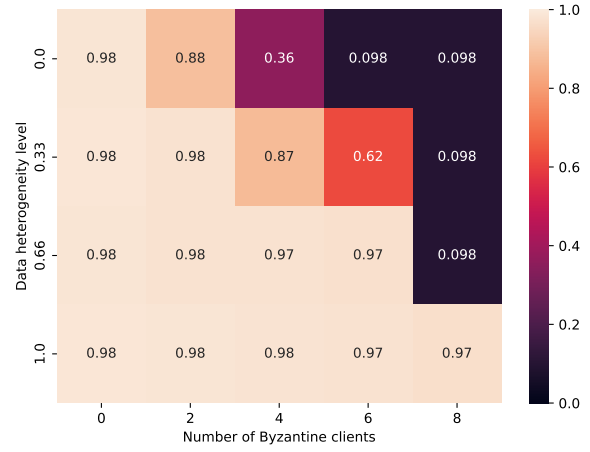


Figure 1: Worst-case test accuracy comparison using the best overall aggregator selection.

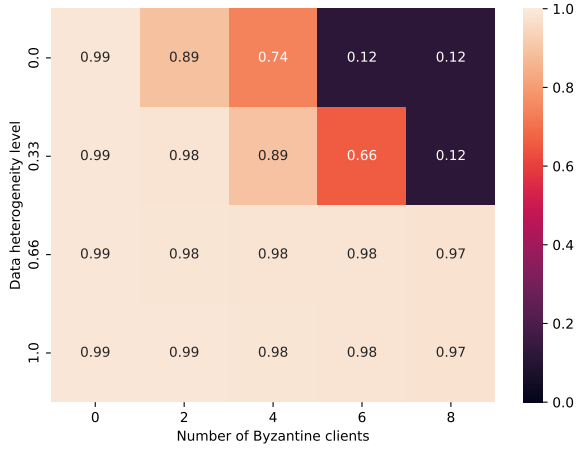
2.2 Centered Clipping



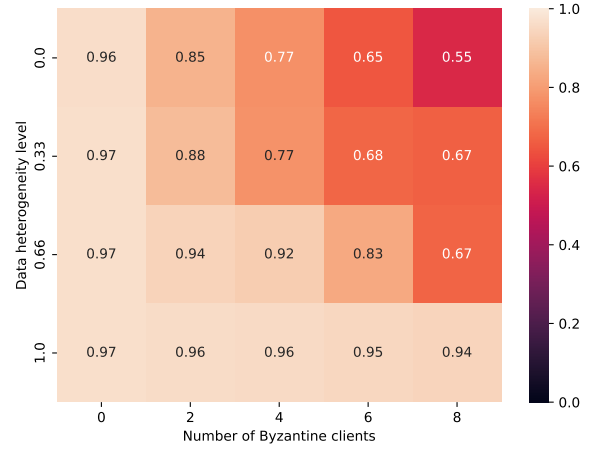
(a) CNN



(b) SNN Rate



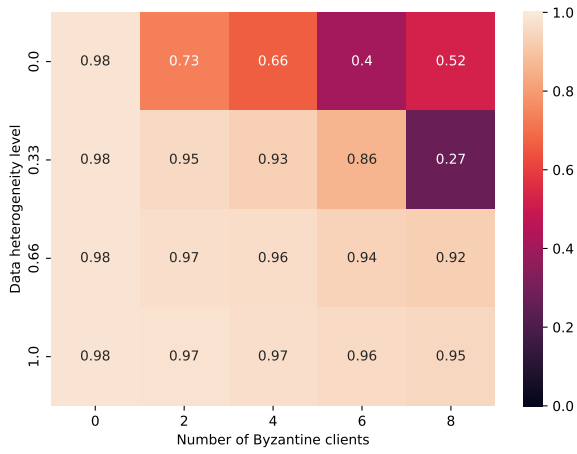
(c) SNN Direct



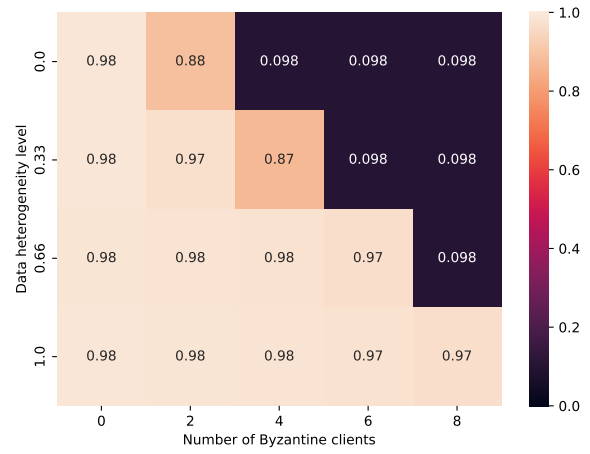
(d) SNN MNIST

Figure 2: Worst-case test accuracy comparison using Centered Clipping.

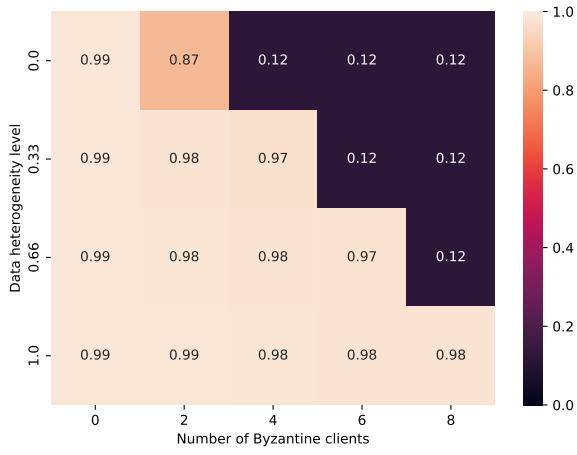
2.3 Geometric Median



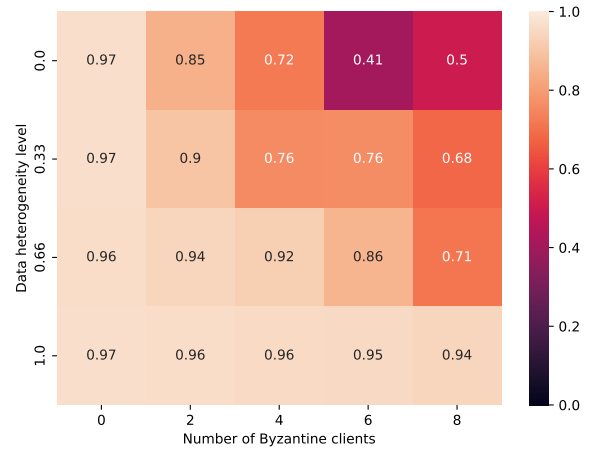
(a) CNN



(b) SNN Rate



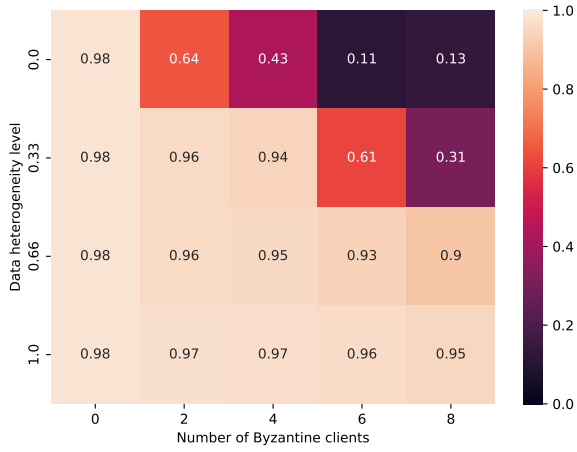
(c) SNN Direct



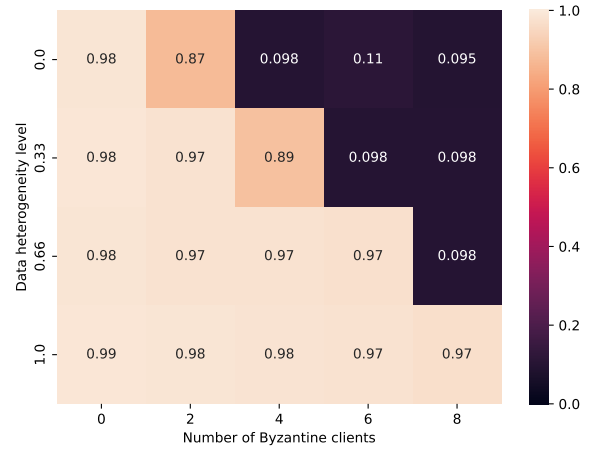
(d) SNN NMNIST

Figure 3: Worst-case test accuracy comparison using Geometric Median.

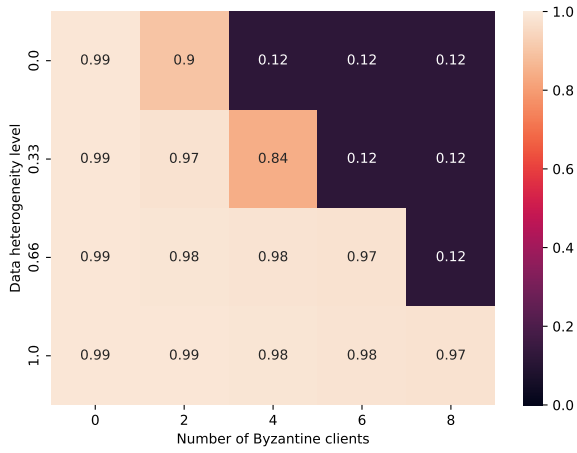
2.4 Multi-Krum



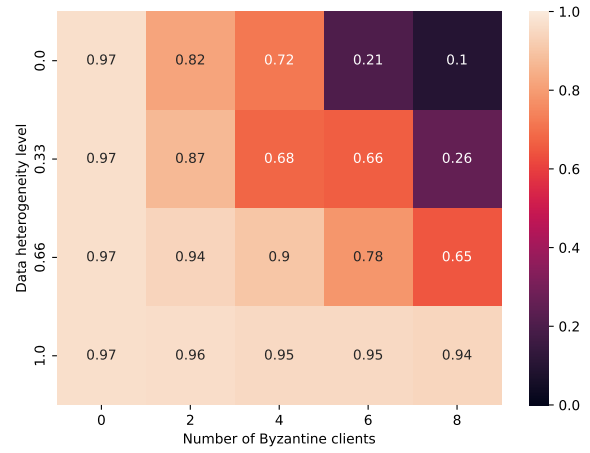
(a) CNN



(b) SNN Rate



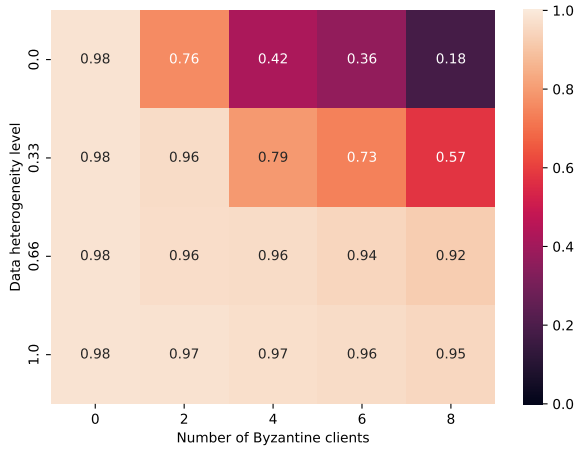
(c) SNN Direct



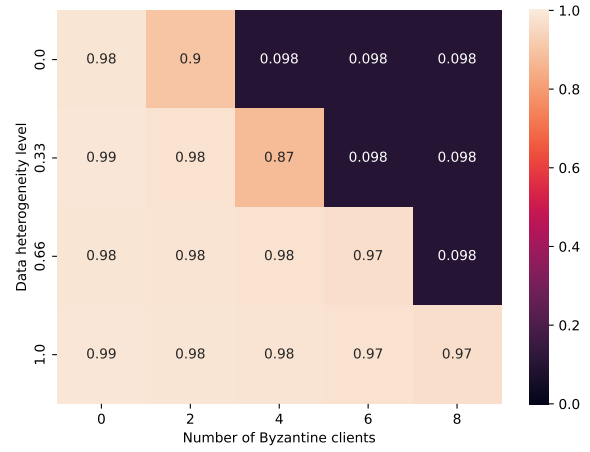
(d) SNN NMNIST

Figure 4: Worst-case test accuracy comparison using Multi-Krum.

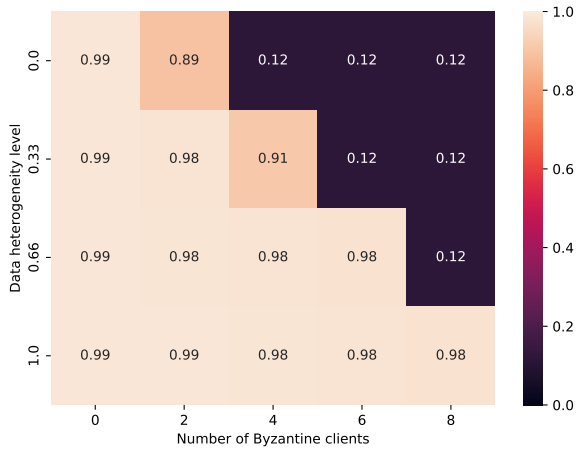
2.5 Trimmed Mean



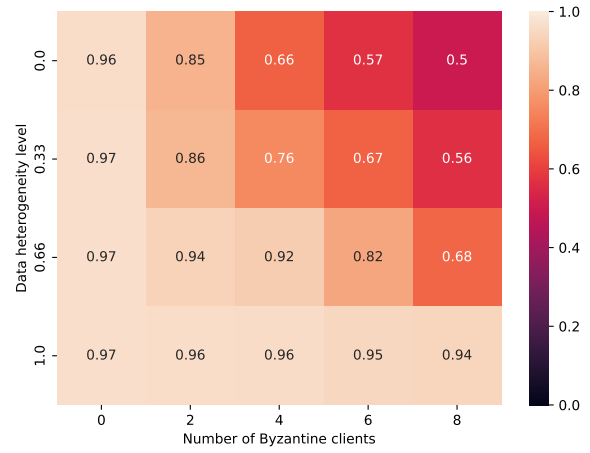
(a) CNN



(b) SNN Rate



(c) SNN Direct



(d) SNN NMNIST

Figure 5: Worst-case test accuracy comparison using Trimmed Mean.

3 Fixed Attack: Optimal A Little Is Enough (ALIE)

This section compares accuracy under the Optimal ALIE Byzantine attack.

3.1 Best Aggregator Choice

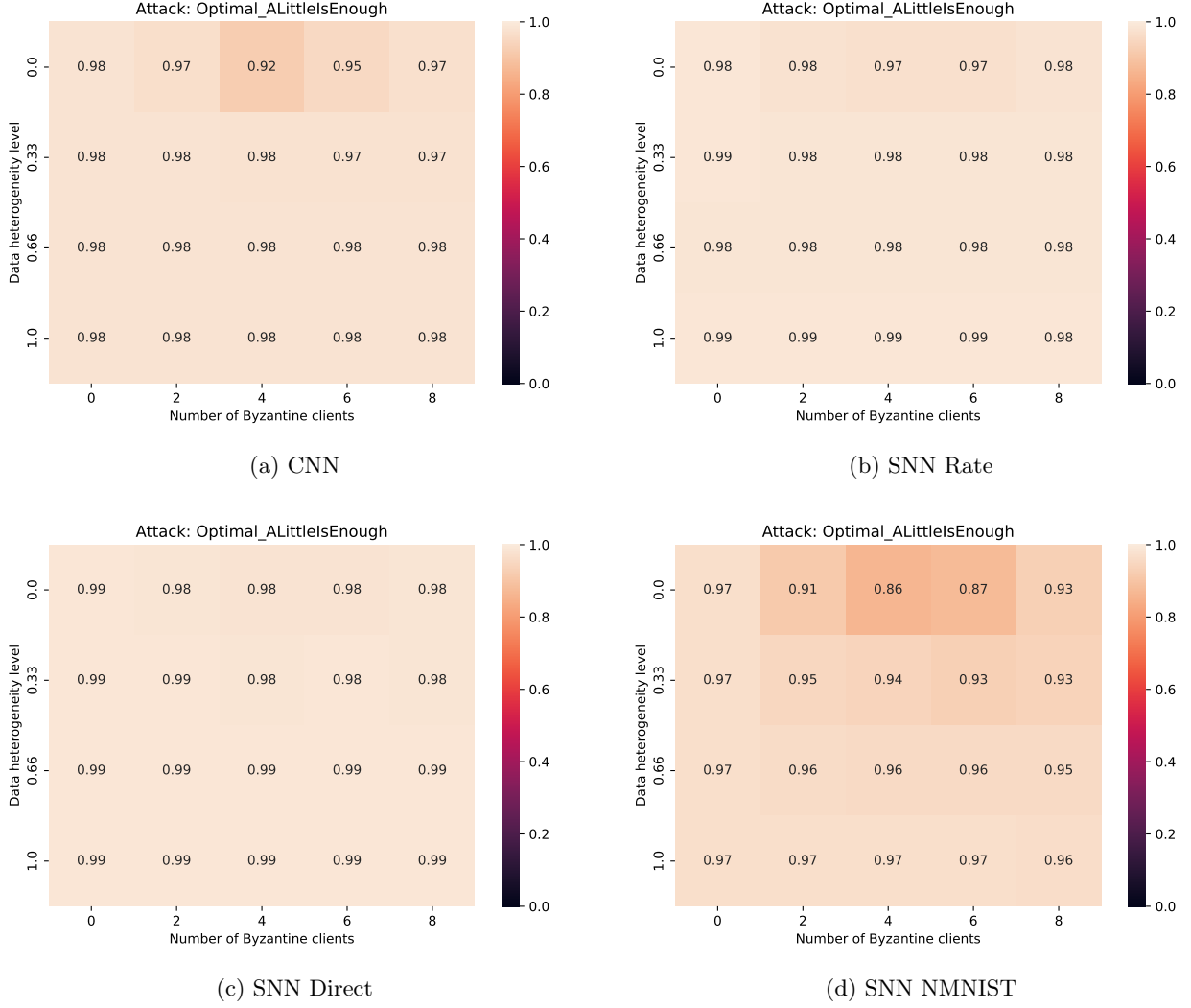
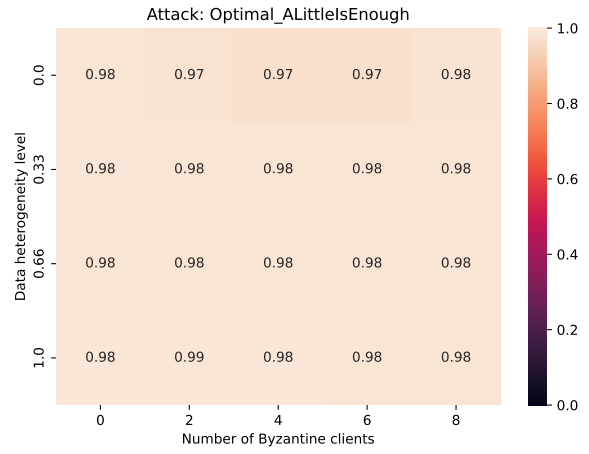


Figure 6: Accuracy comparison under Optimal ALIE attack using the best aggregator selection.

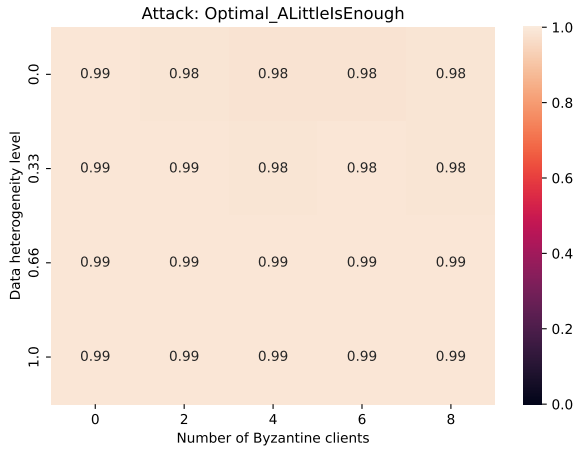
3.2 Centered Clipping



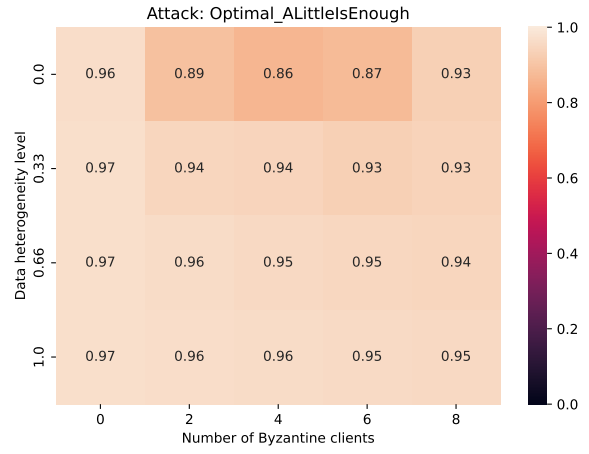
(a) CNN



(b) SNN Rate



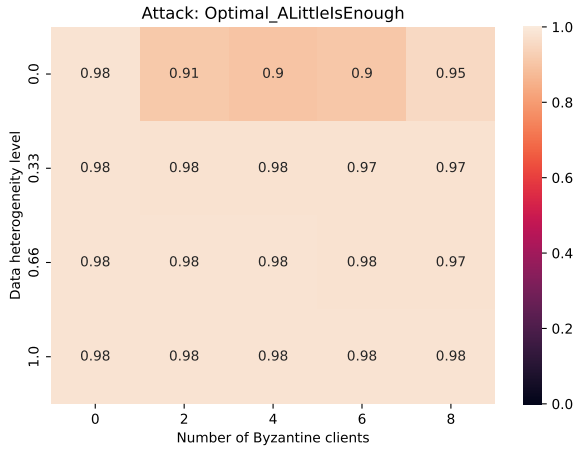
(c) SNN Direct



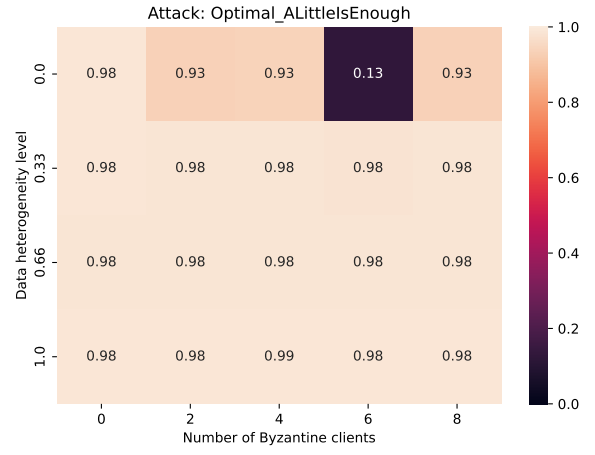
(d) SNN MNIST

Figure 7: Accuracy comparison under Optimal ALIE attack using Centered Clipping.

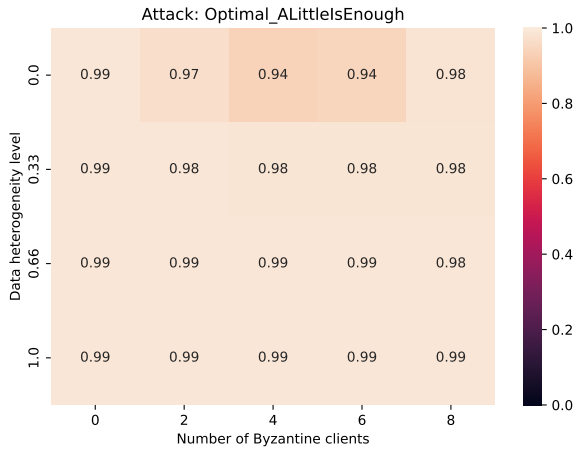
3.3 Geometric Median



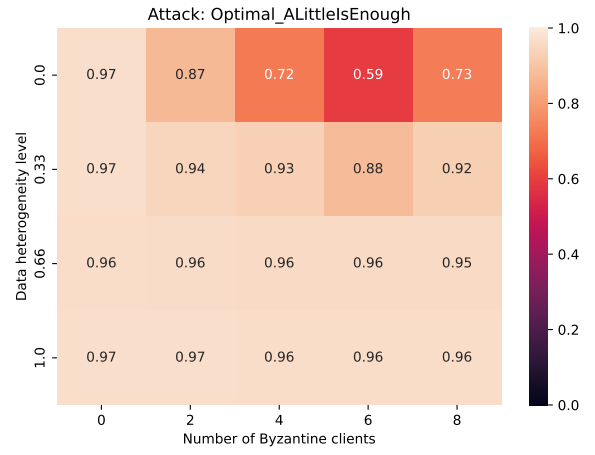
(a) CNN



(b) SNN Rate



(c) SNN Direct



(d) SNN NMNIST

Figure 8: Accuracy comparison under Optimal ALIE attack using Geometric Median.

3.4 Multi-Krum

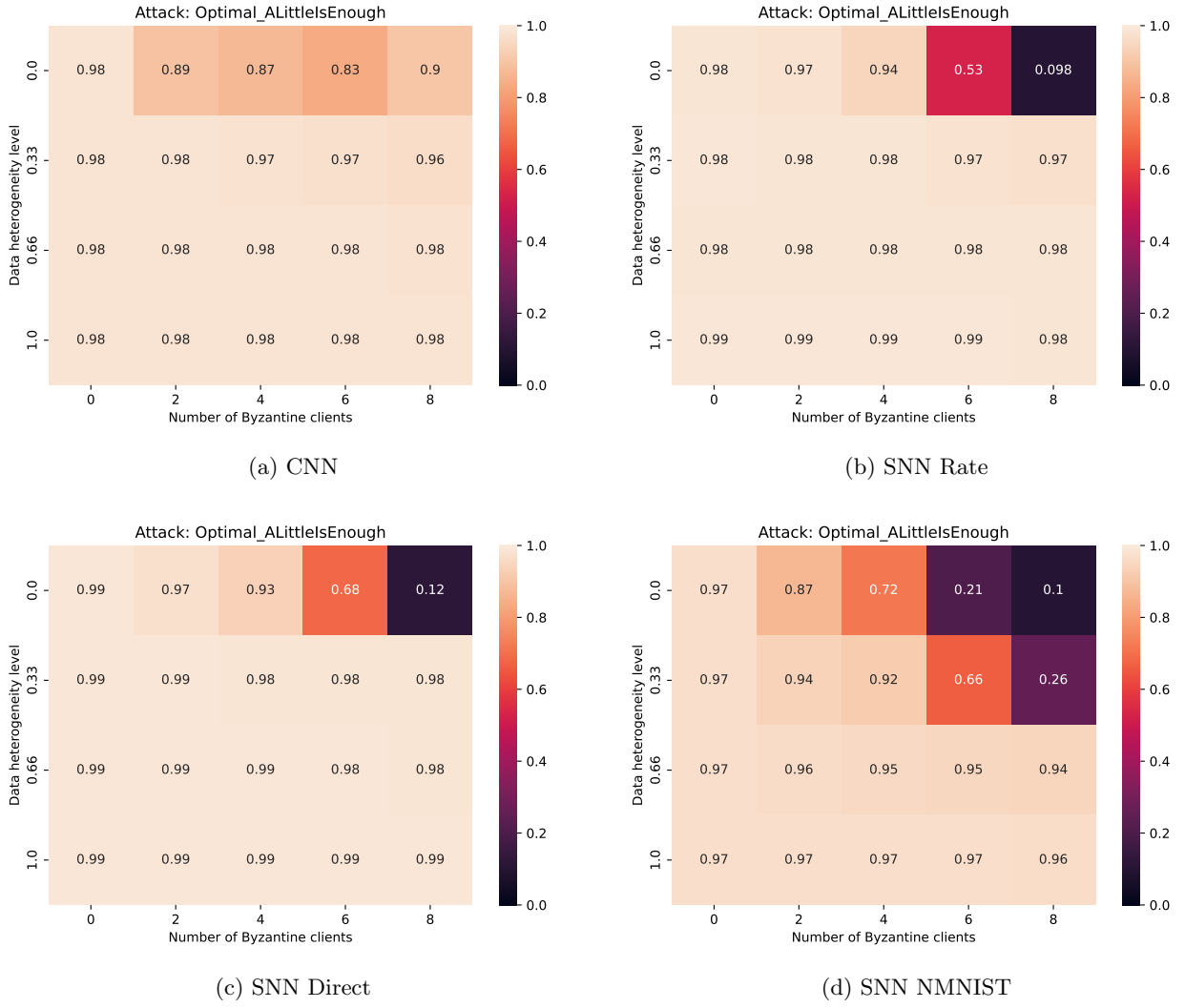
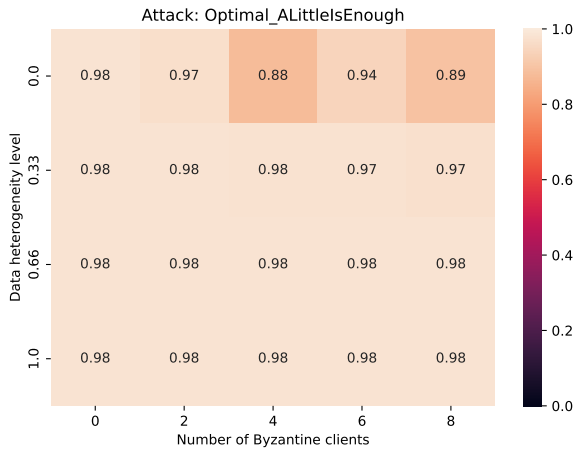
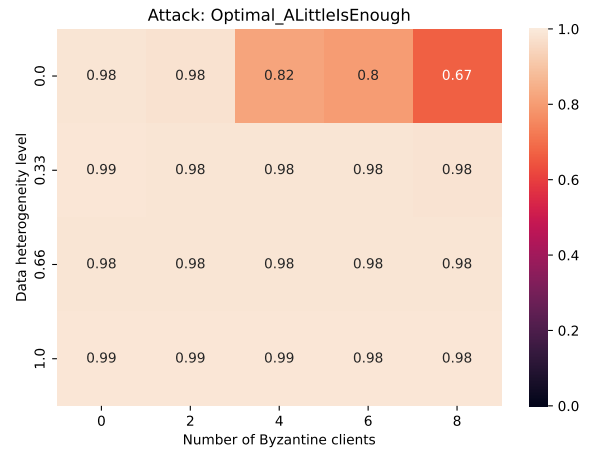


Figure 9: Accuracy comparison under Optimal ALIE attack using Multi-Krum.

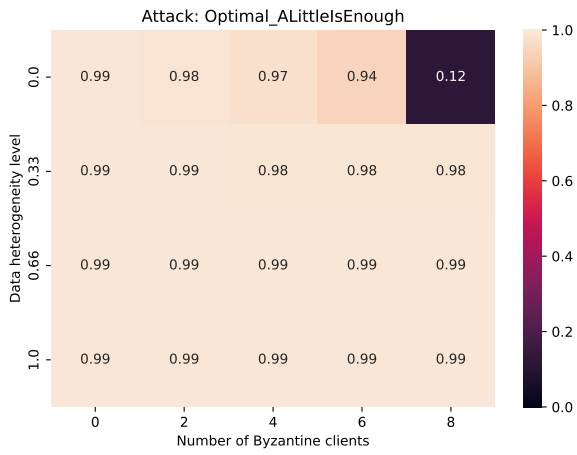
3.5 Trimmed Mean



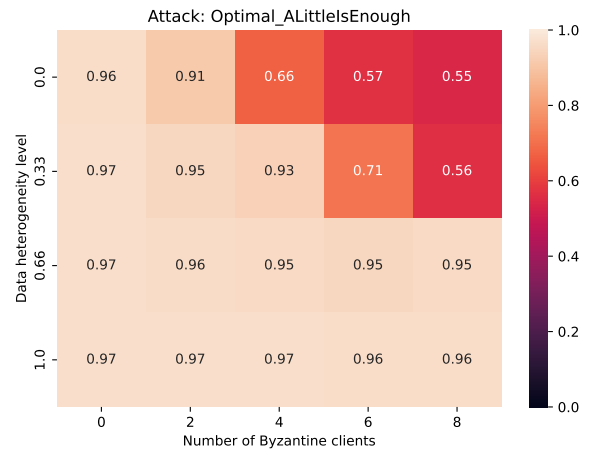
(a) CNN



(b) SNN Rate



(c) SNN Direct



(d) SNN NMNIST

Figure 10: Accuracy comparison under Optimal ALIE attack using Trimmed Mean.

4 Fixed Attack: Optimal Inner Product Manipulation (IPM)

This section compares accuracy under the Optimal IPM Byzantine attack.

4.1 Best Aggregator Choice

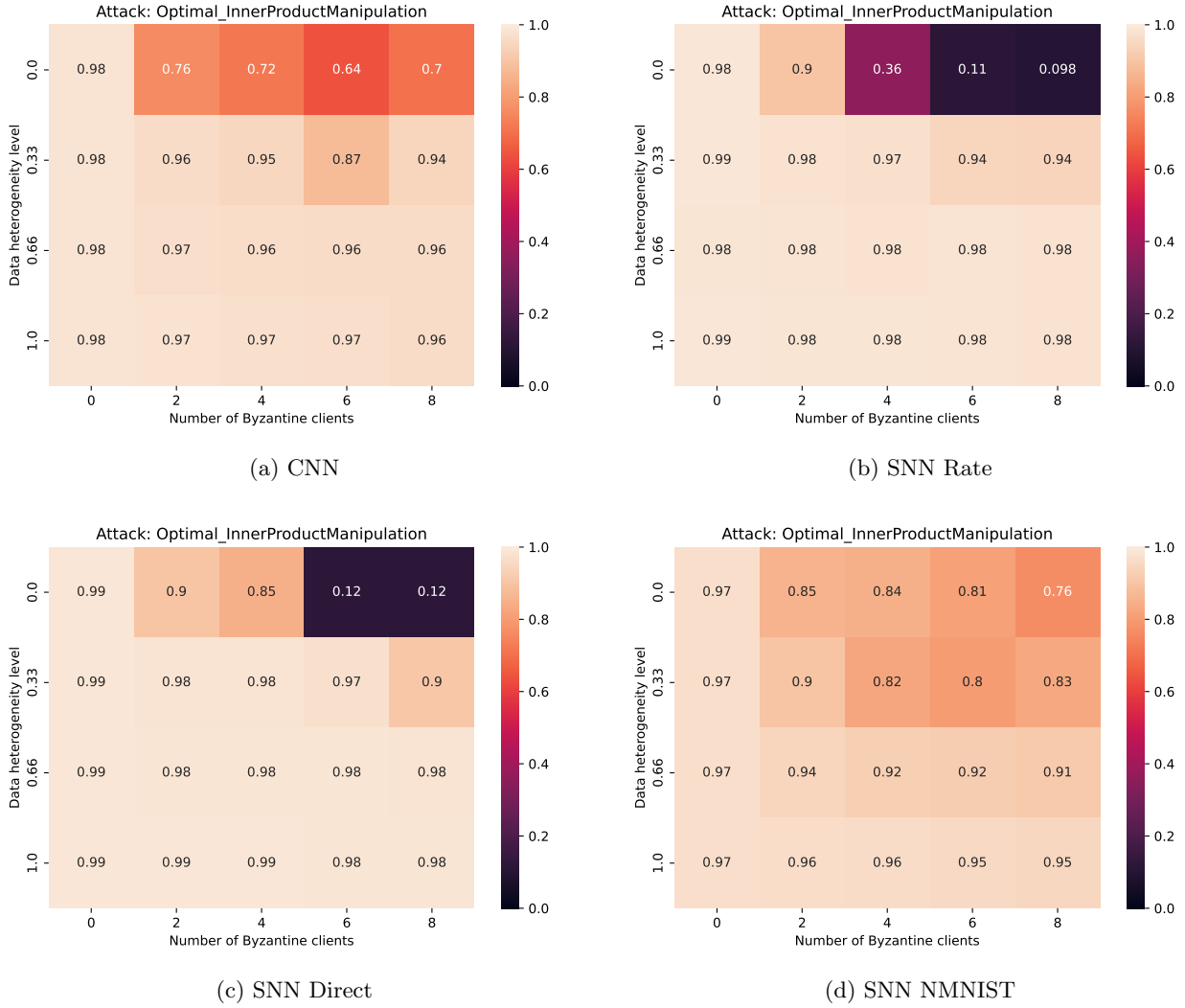
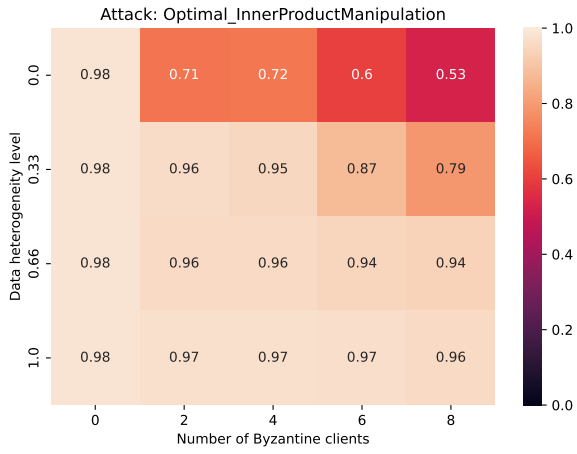
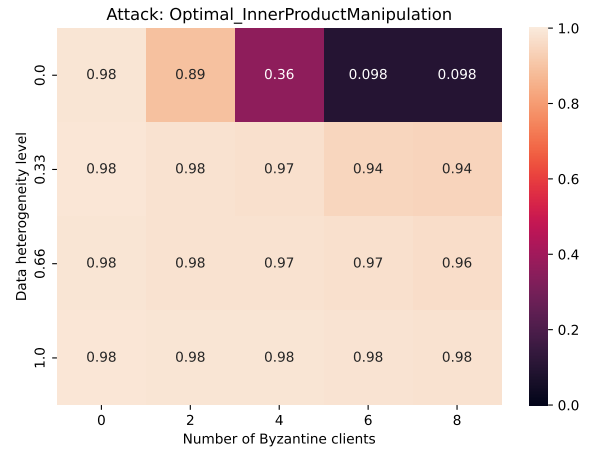


Figure 11: Accuracy comparison under Optimal IPM attack using the best aggregator selection.

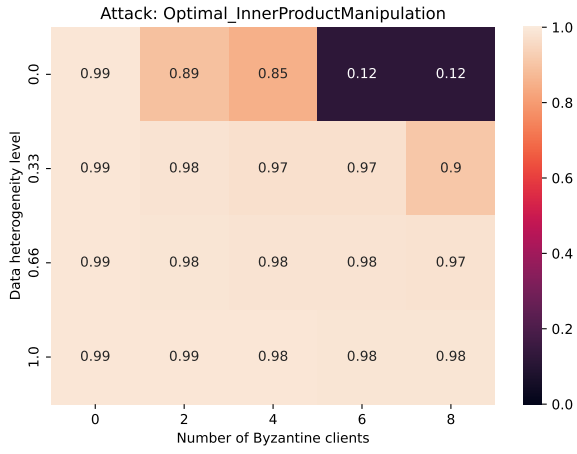
4.2 Centered Clipping



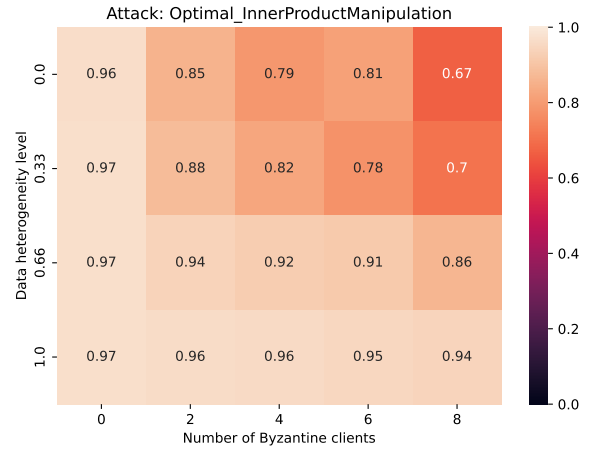
(a) CNN



(b) SNN Rate



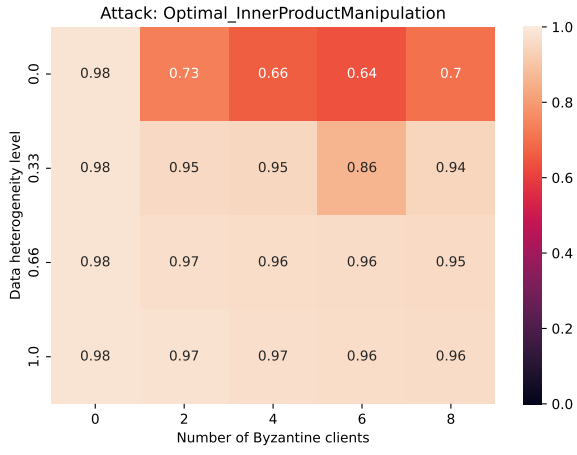
(c) SNN Direct



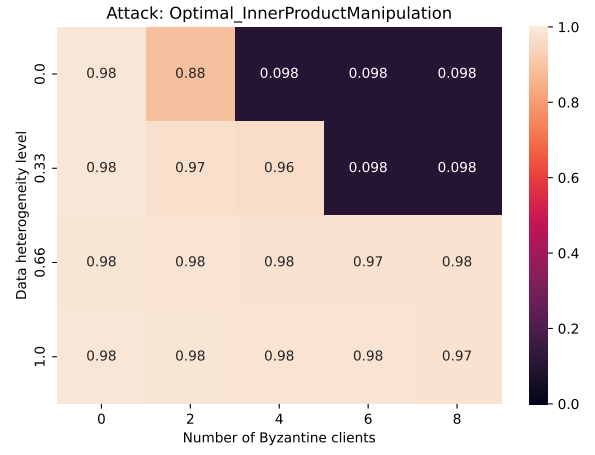
(d) SNN NMNIST

Figure 12: Accuracy comparison under Optimal IPM attack using Centered Clipping.

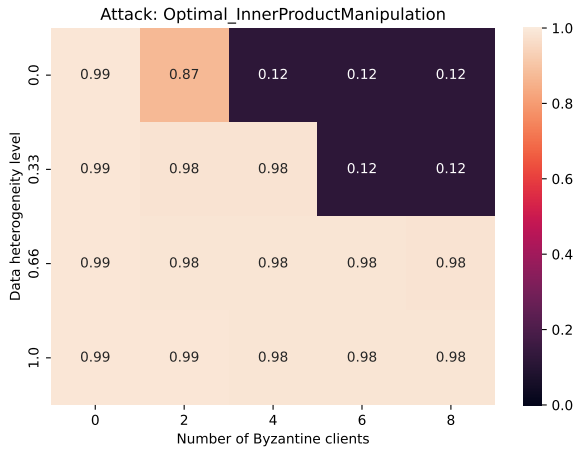
4.3 Geometric Median



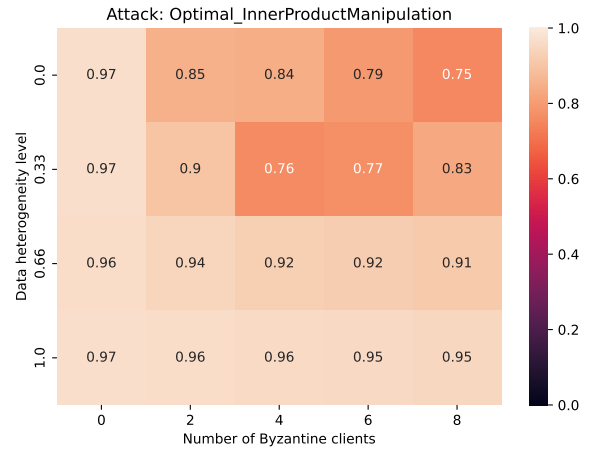
(a) CNN



(b) SNN Rate



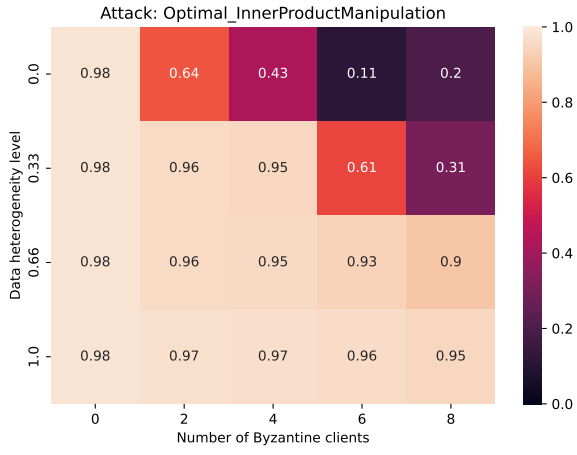
(c) SNN Direct



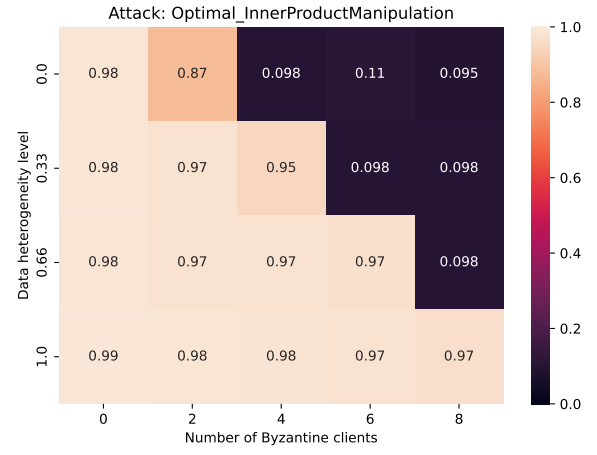
(d) SNN NMNIST

Figure 13: Accuracy comparison under Optimal IPM attack using Geometric Median.

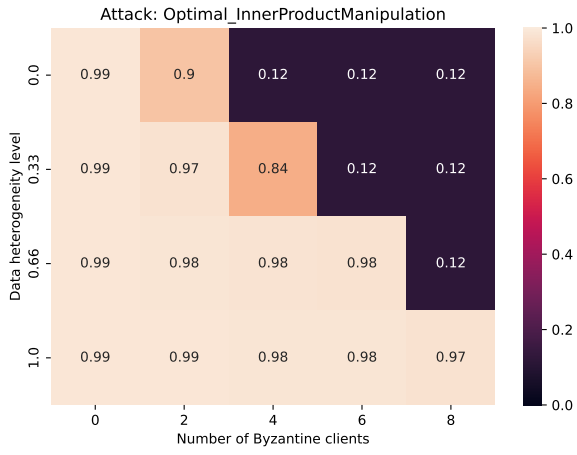
4.4 Multi-Krum



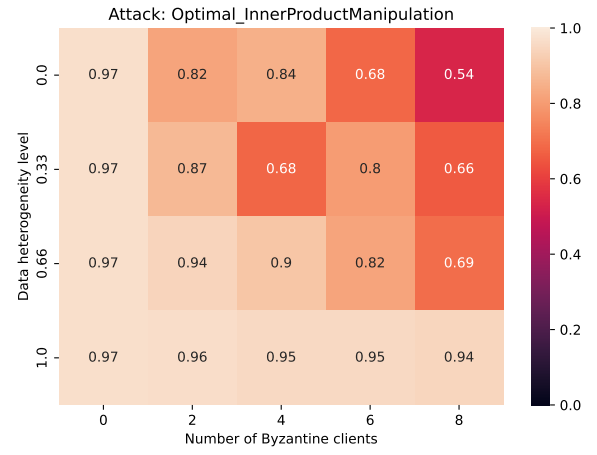
(a) CNN



(b) SNN Rate



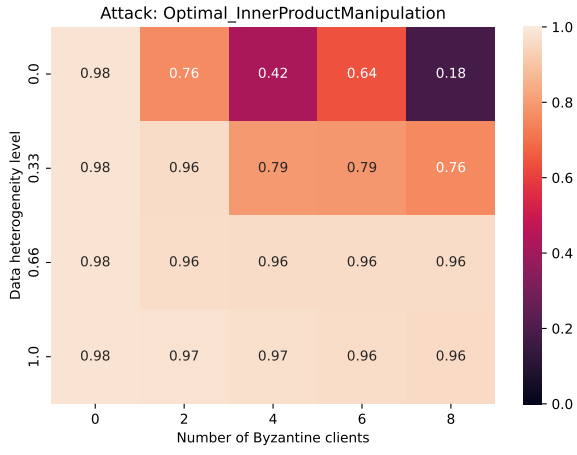
(c) SNN Direct



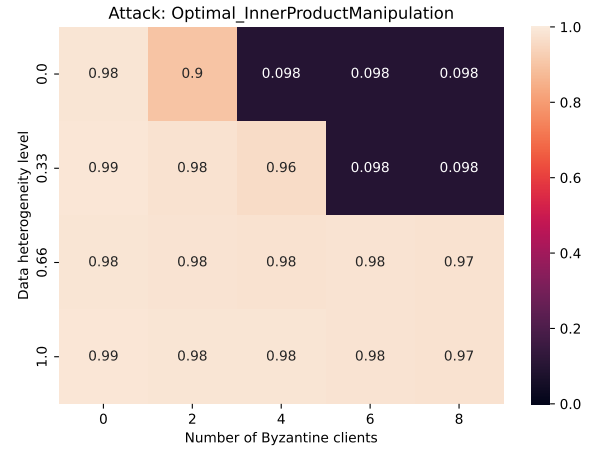
(d) SNN NMNIST

Figure 14: Accuracy comparison under Optimal IPM attack using Multi-Krum.

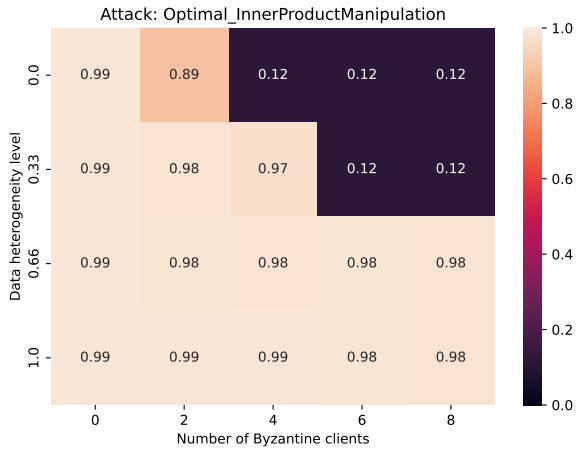
4.5 Trimmed Mean



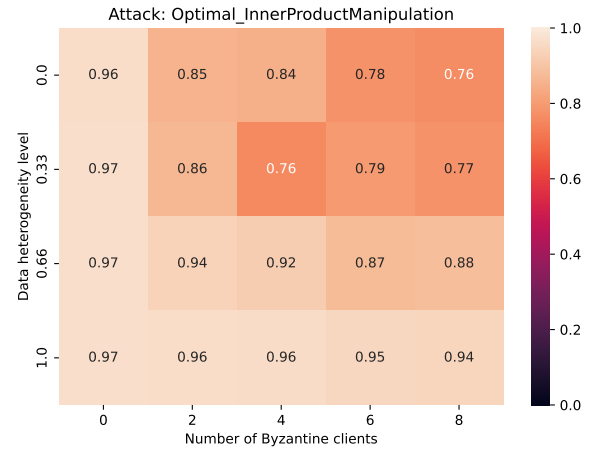
(a) CNN



(b) SNN Rate



(c) SNN Direct



(d) SNN NMNIST

Figure 15: Accuracy comparison under Optimal IPM attack using Trimmed Mean.

5 Fixed Attack: Sign Flipping

This section compares accuracy under the Sign Flipping Byzantine attack.

5.1 Best Aggregator Choice

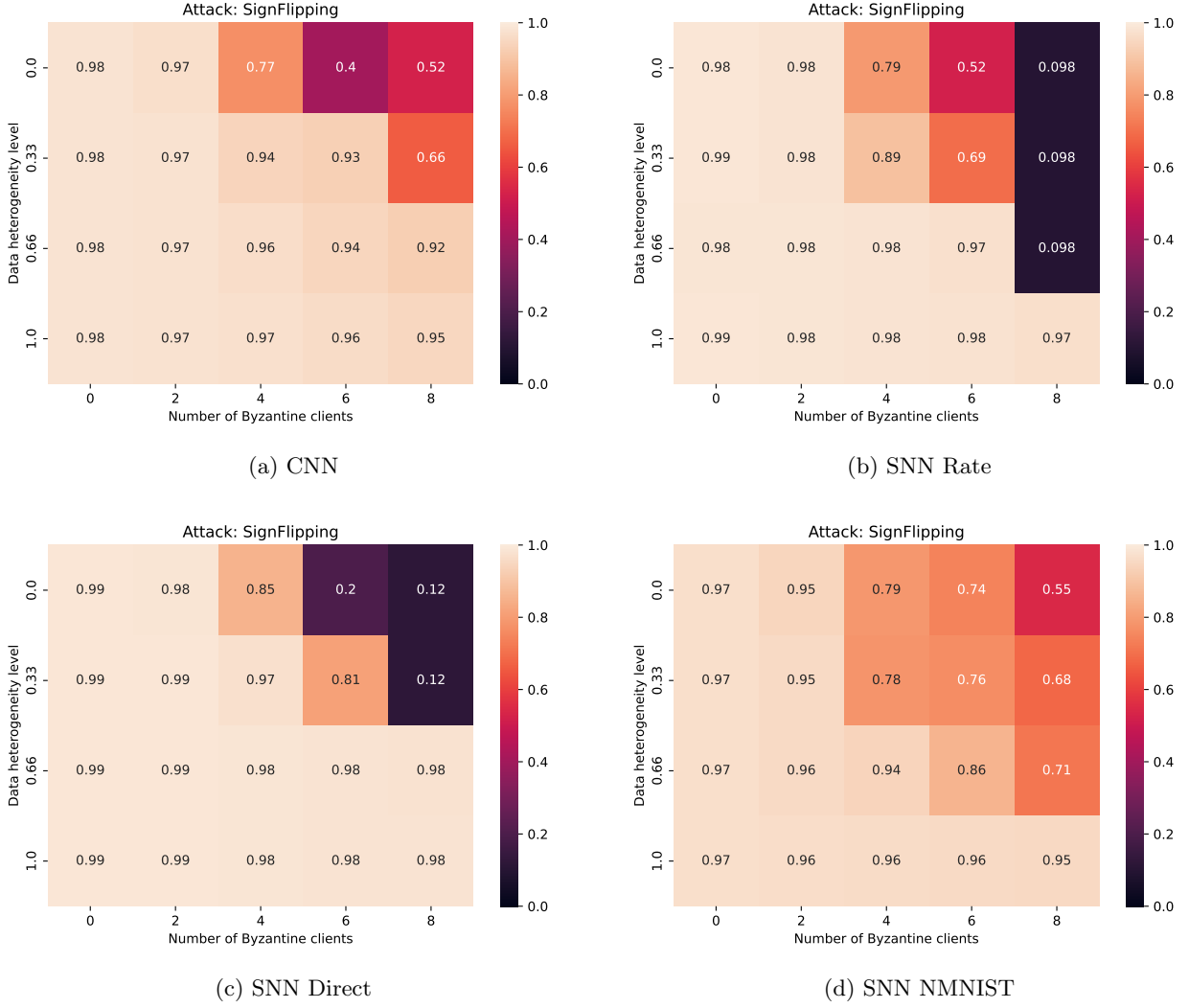


Figure 16: Accuracy comparison under Sign Flipping attack using the best aggregator selection.

5.2 Centered Clipping

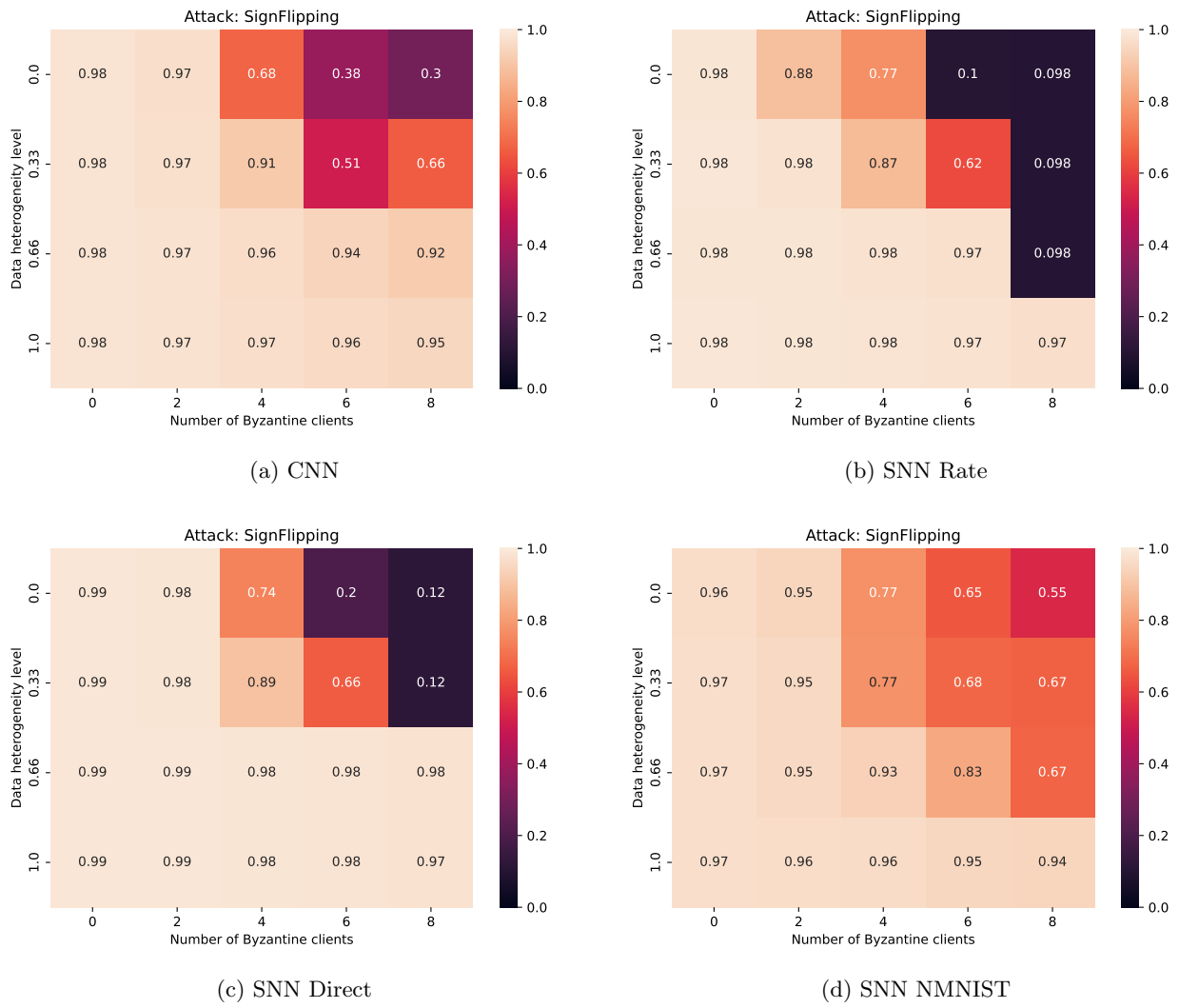
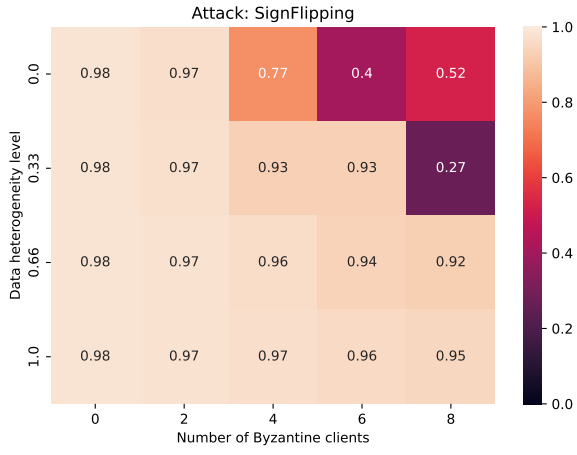
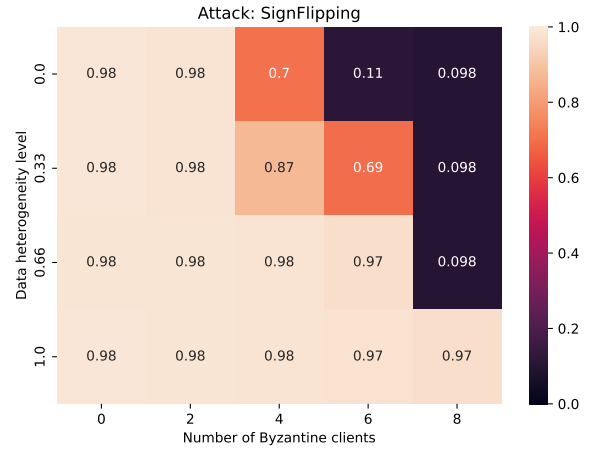


Figure 17: Accuracy comparison under Sign Flipping attack using Centered Clipping.

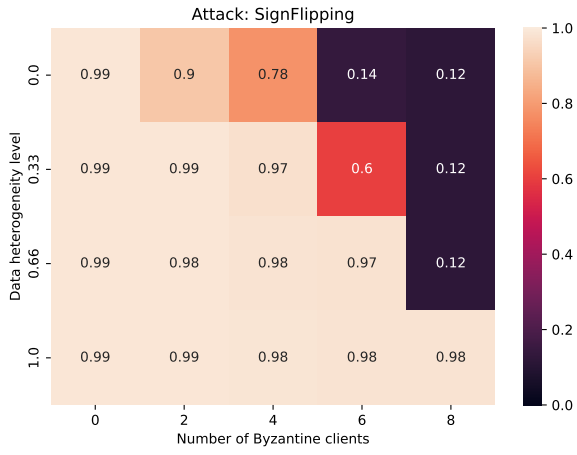
5.3 Geometric Median



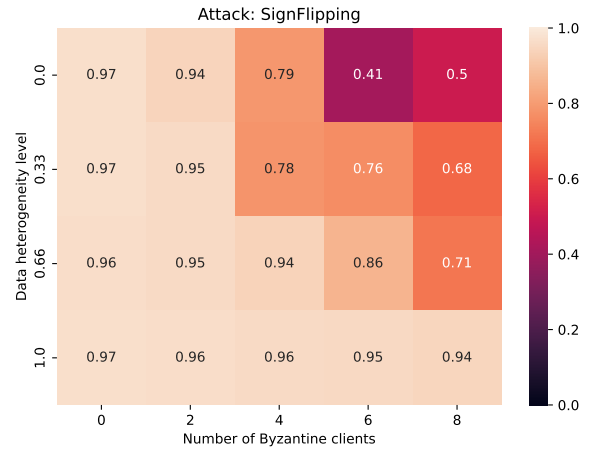
(a) CNN



(b) SNN Rate



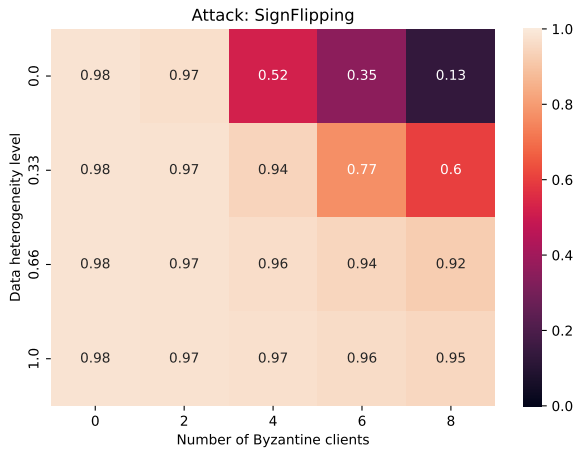
(c) SNN Direct



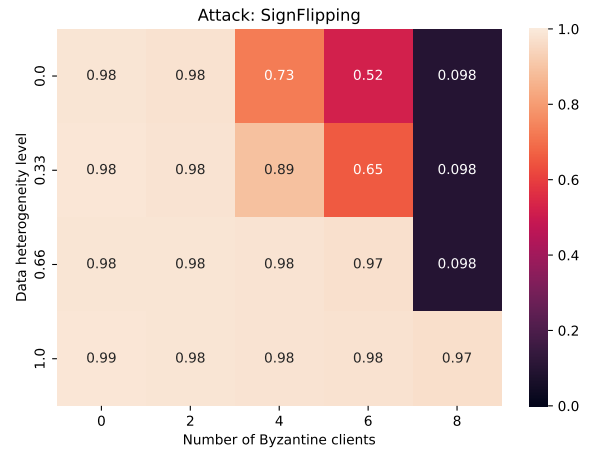
(d) SNN NMNIST

Figure 18: Accuracy comparison under Sign Flipping attack using Geometric Median.

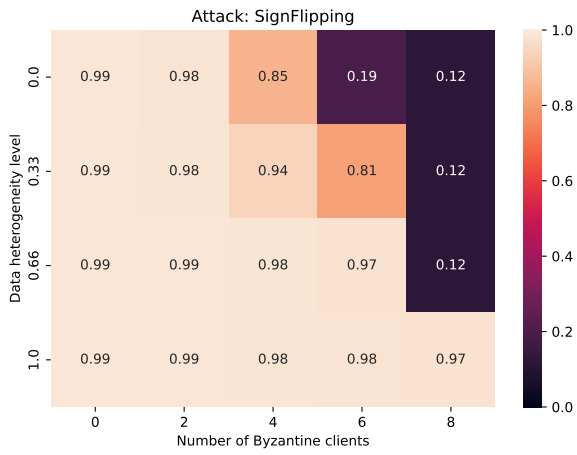
5.4 Multi-Krum



(a) CNN



(b) SNN Rate



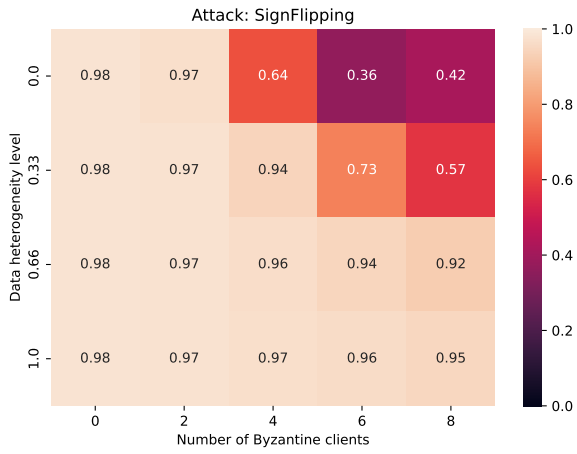
(c) SNN Direct



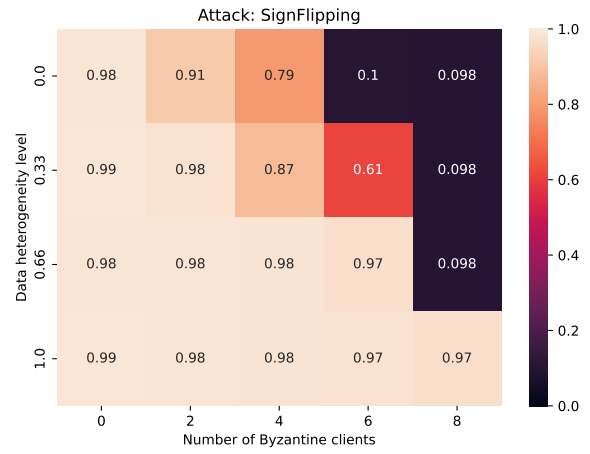
(d) SNN NMNIST

Figure 19: Accuracy comparison under Sign Flipping attack using Multi-Krum.

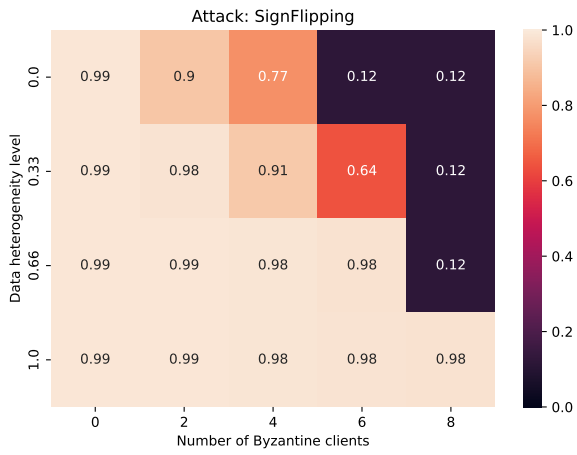
5.5 Trimmed Mean



(a) CNN



(b) SNN Rate



(c) SNN Direct



(d) SNN NMNIST

Figure 20: Accuracy comparison under Sign Flipping attack using Trimmed Mean.